

AI5707 – Pattern Recognition and Machine Learning

Assignment 4: Multivariate Gaussian Density functions, Covariance Matrices, Constant Density Curves

Due: Sept 22 2026

1 Objective

In this assignment, you will learn to generate data from 2-D Gaussian density functions with a variety of Covariance matrices.

a. $C = \sigma^2 I$

b. $C = \begin{bmatrix} \sigma_{11}^2 & 0 \\ 0 & \sigma_{22}^2 \end{bmatrix}$

c. $\mathbf{C} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12} \\ \sigma_{21} & \sigma_{22}^2 \end{bmatrix}$

2 Problems

1. Isotropic Covariance Matrix:

$$\mathbf{C} = \sigma^2 \mathbf{I}$$

- a. Generate two datasets (D_1, D_2) Gaussians with μ_1, μ_2 and $\sigma^2 = 1$ as in the first assignment.
- b. Use data set D_1 to correspond to one dimension and D_2 to correspond to the other dimension. Now you will have the vector of values $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and the mean vector $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$.
- c. Estimate the covariance matrix from the data, compute eigenvalues and eigenvectors.
- d. Plot the constant density curves defined by $(\mathbf{x} - \boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$
- e. Estimate eigenvalues and eigenvectors. Plot the eigenvectors, mark the length of major and minor axes.
- f. Compare with the eigenvalues obtained.

2. Diagonal Covariance Matrix

$$\mathbf{C} = \begin{bmatrix} \sigma_{11}^2 & 0 \\ 0 & \sigma_{22}^2 \end{bmatrix}$$

- a. Generate data corresponding to the changed covariance matrix $\mathbf{C}$ as:

$$\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}(\mathbf{X} - \boldsymbol{\mu}) \quad (1)$$

- b. Repeat the previous task.

3. Full Covariance Matrix

$$\mathbf{C} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12} \\ \sigma_{21} & \sigma_{22}^2 \end{bmatrix}$$

- a. Generate data as in the previous question.
- b. Plot the curves as before.
4. a. Generate two different datasets $\mathbf{Y}_1$ & $\mathbf{Y}_2$ corresponding to classes ω_1 & ω_2 where each class is represented by $\mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$.
- b. What can you conclude about the decision boundaries between the two classes ω_1 & ω_2 ? Experiment with different values of $\boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma}_i$